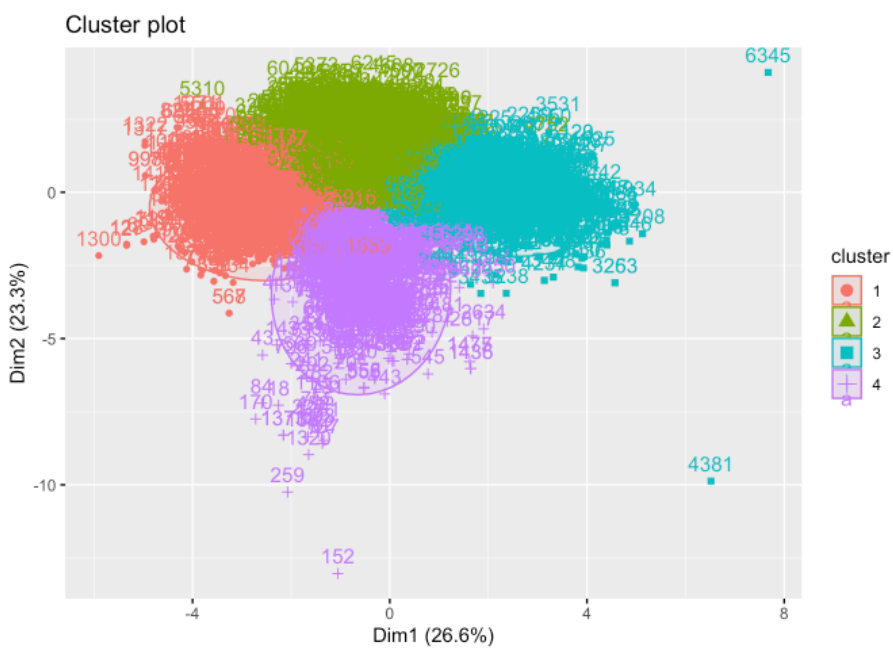
Case
Introduction
Get all the required libraries
View Data
Clustering
How many clusters are optimal?
Create and Visualize clusters
Visualize Cluster Variables
View PCA's
Predicitive Model
Decision Tree
Distance to Center Cluster
Final Descrition of Cluster Characterizations
Compare the two methods
Conclusion

```
# We can clearly see four distinct regions of the plot, however, not all are homogeneous. This may indi
cate the the four main types of wine may share overlapping qualities with other types. However, this he
terogeneity is less than it was with seed 1-5.
```

Visualize Via ellipses

```
fviz_cluster(km_1, data = wine_data, ellipse = TRUE, ellipse.type = "norm", ellipse.level = .95, ellips
e.alpha = .1)
```




```
#visualization with ellipses is nearly identical, we still see some heterogeneity, and lack of dispersi
on between the clusters.
```